

Learning in a Custom 2D Platformer Environment

CS258: Introduction to
Reinforcement Learning

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Motivation

- **Gridworld, Frogger, and Simple Hoppers Explored in class**
- **Hoped to explore a more dynamic, noisy environment**
- **Emulated “Junimo Kart” which is an arcade style game within “Stardew Valley”**
 - **RL agent would have to learn timing and jump patterns**
 - **Goal was to bridge class concepts and real game behavior**

Gym Environment: Observation Space

Constructed Gym environment from scratch, allowing our agent to see

- 5 channel tile map, with mappings (C, H, W)

- Platforms
- Boulders (standable)
- Roadblocks (bounceable)
- Coins
- Fruits



- Player state, no mappings, direct floats

- Y position, Y velocity, on_ground



Gym Environment: Action Space / Physics

Action Space (0, 1)

- **0: Do nothing**
 - On ground: resets jump_hold
- **1: Jump**
 - On ground: sets velocity to jump strength (4.0) & jump_hold to 1
 - Off ground: Add a small jump boost (0.3) to velocity
 - Unless it reached jump_max_hold (7)

Gravity added (-0.7) to velocity

- Reset later if on ground and not pit

```
# Jump Logic
self.allow_jump_after -= 1
can_jump = (self.allow_jump_after <= 0)
if action == 1 and can_jump:
    if self._is_on_ground() or self._is_on_boulder(): # Start Jump
        self.y_vel = self.jump_strength
        self.jump_hold = 1

    else: # Hold jump if not on ground
        if 0 <= self.jump_hold < self.jump_max_hold:
            # Continue jump if still holding jump
            self.y_vel += self.jump_boost
            self.jump_hold += 1
        else:
            # Jump button released
            if self._is_on_ground() or self._is_on_boulder(): # End Jump
                self.jump_hold = 0

# Physics
prev_y = self.y
self.y_vel += self.gravity
self.y = max(0.0, self.y + self.y_vel * 0.3)
```

...

```
# Normal, safe Landing and snap up from slope
if self.y <= ground_y:
    self.y = ground_y
    self.y_vel = 0
    self.jump_hold = 0
```

Gym Environment: Object Physics

- **Platforms**

- Kills player if not in slack (2.0) range
- Otherwise snaps player up to ground level
- Implemented for fairness when jumping pits

- **Pits**

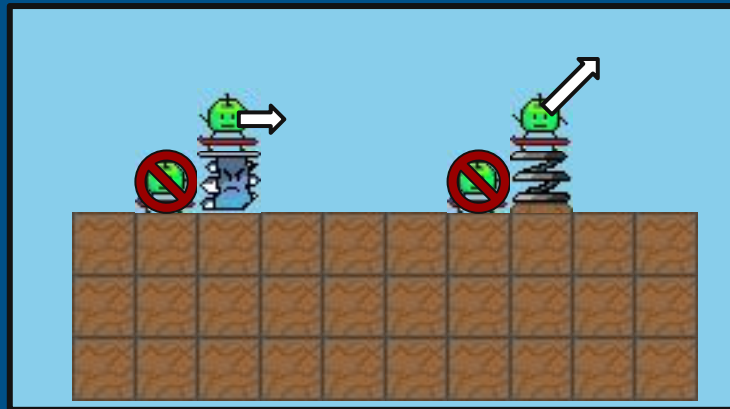
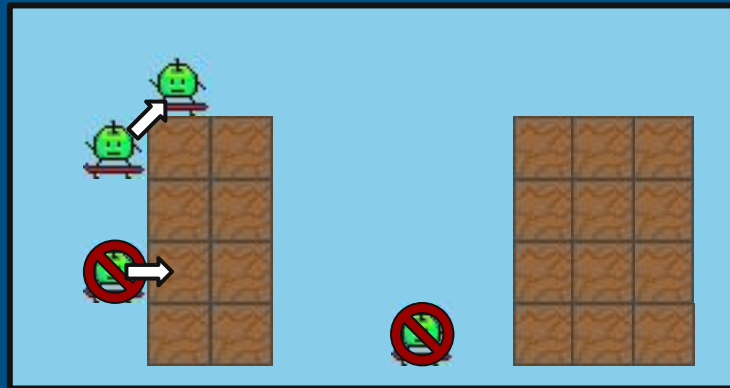
- Kills player under death barrier (0.5 height)

- **Coins / Fruits**

- Collects object (sets grid to 0.0) and adds reward

- **Roadblock / Boulder**

- If object is directly in front, Kills player
- Bounces if on top of roadblock, platform if on top of boulder

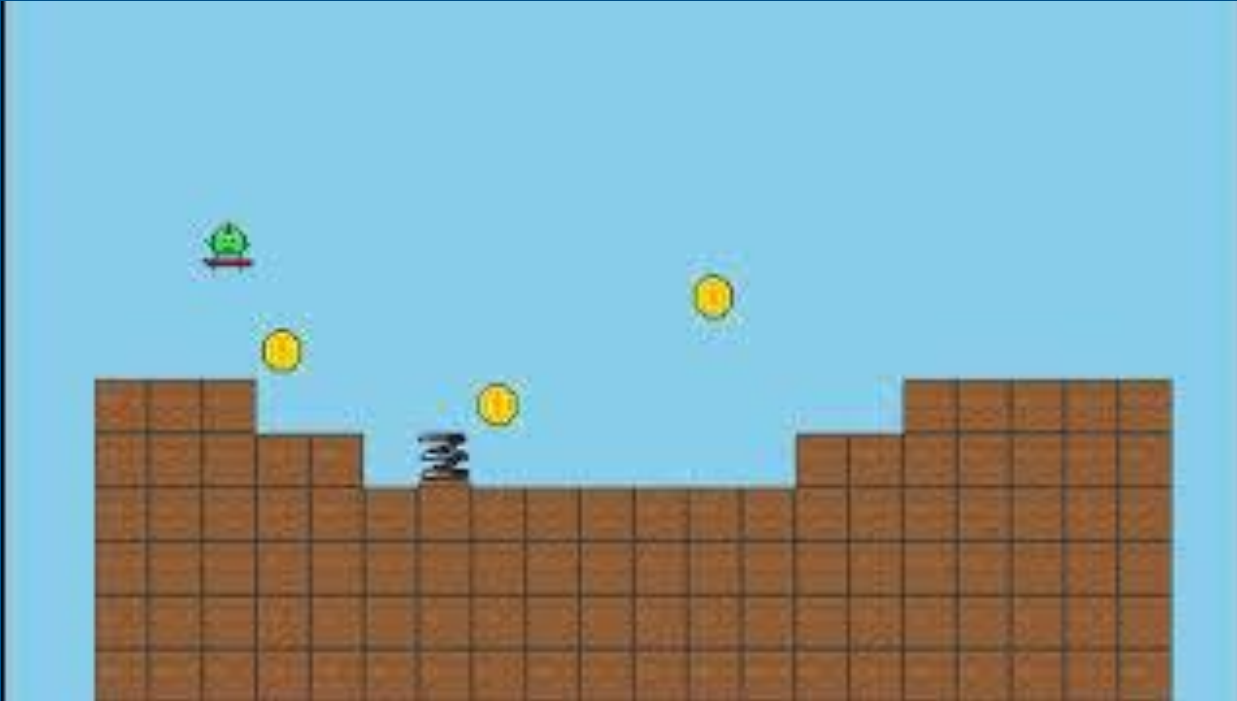


Gym Environment: Object Spawn Mechanics

- **Initial World Generation makes safe spawn**
 - No Pits, No Obstacles
 - Exactly 3 fruits spawn columns chosen
- **Ground columns spawns after scrolling**
 - Logic for Pit generation (pit cooldown / length), as well as terrain variation
- **Objects are spawned**
 - Prevents objects from spawning in the same column as each other
 - Provides probabilities of objects spawning
 - Prevents obstacles from spawning near fruit (radius 5)

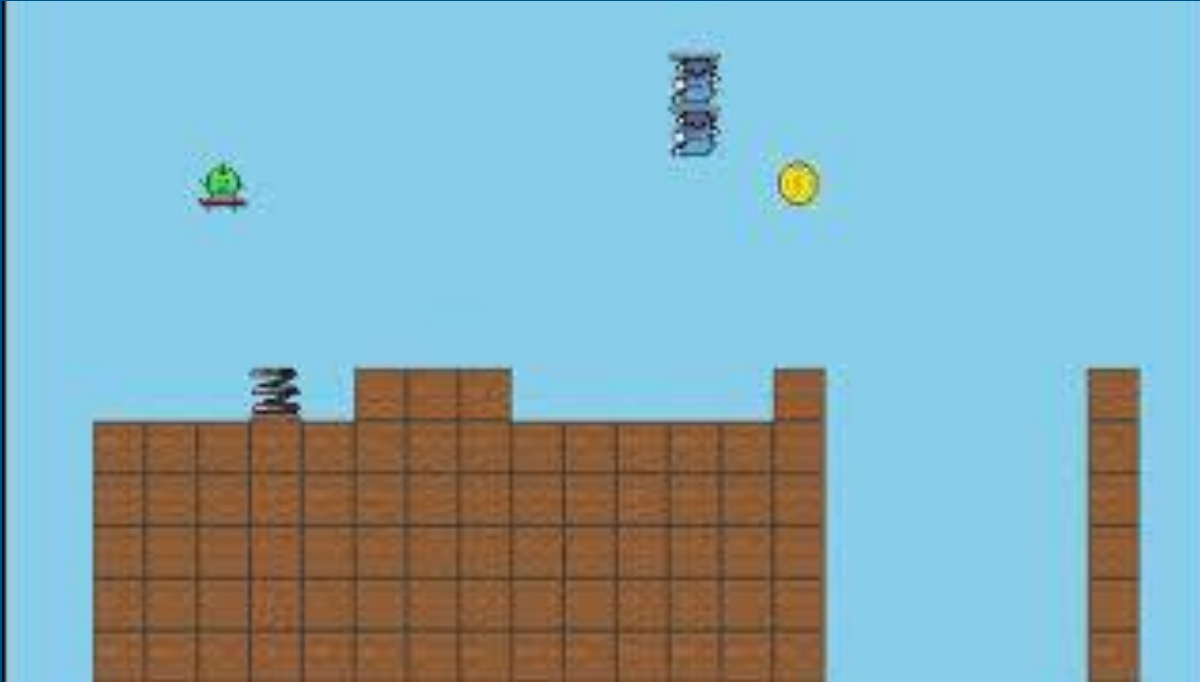
Video Example Result

- Average Result from random inputs on the environment



Video Example Result

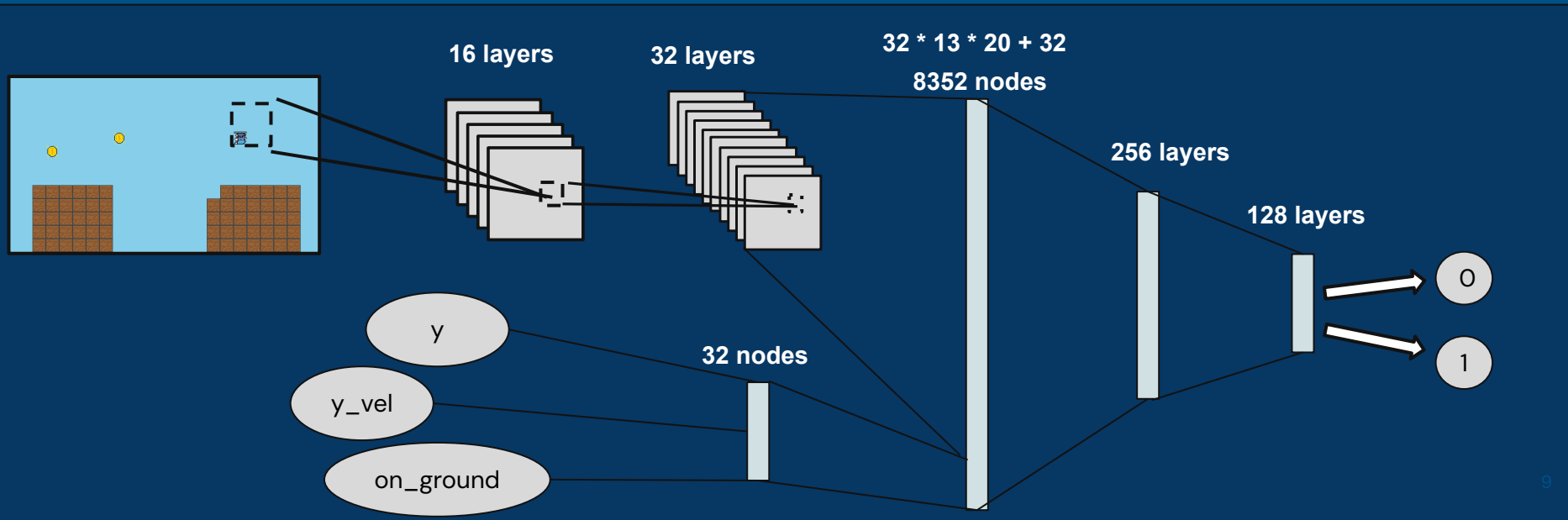
- Random good run from random input



Model

Adapted PPO Implementation

- Policy and Value networks process tiles and player state together
- Curriculum Learning extends time limit over training steps

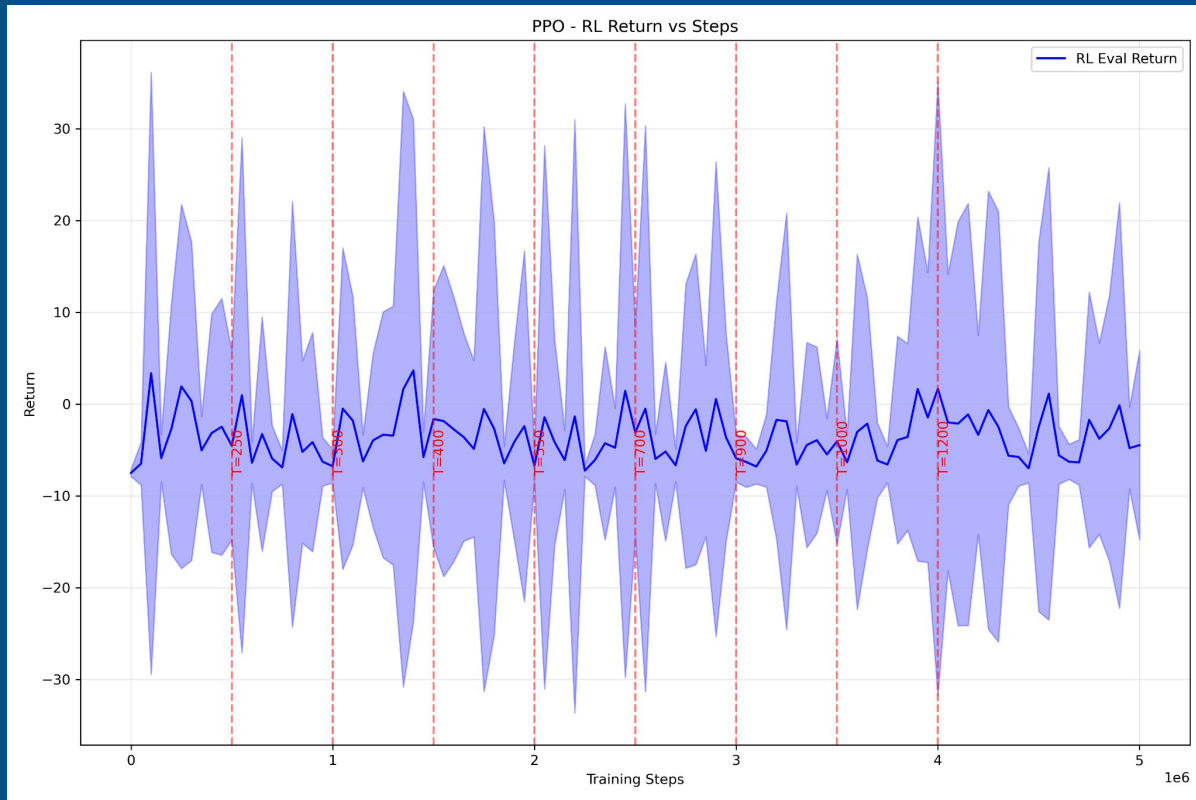


Evaluation Results and Challenges

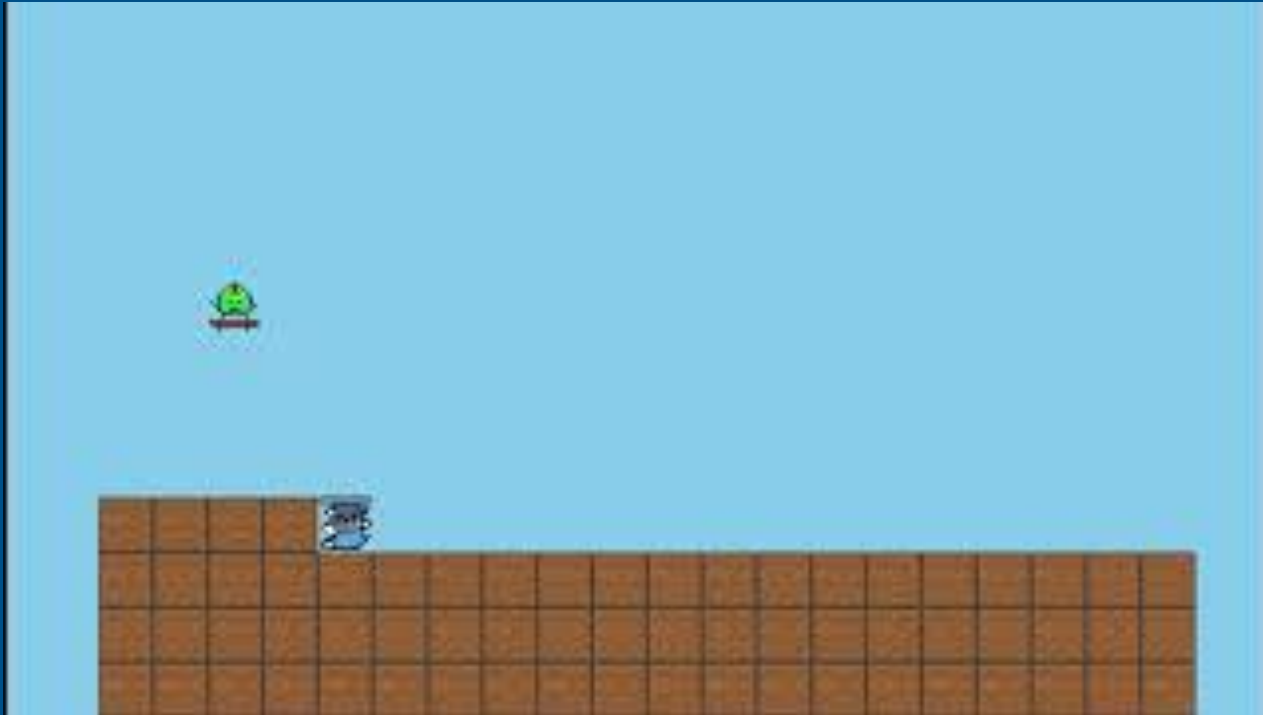
Trained model for 5 M steps,
evaluated every 50,000 steps.

The results are unfortunately
not very positive, which may
be attributed to:

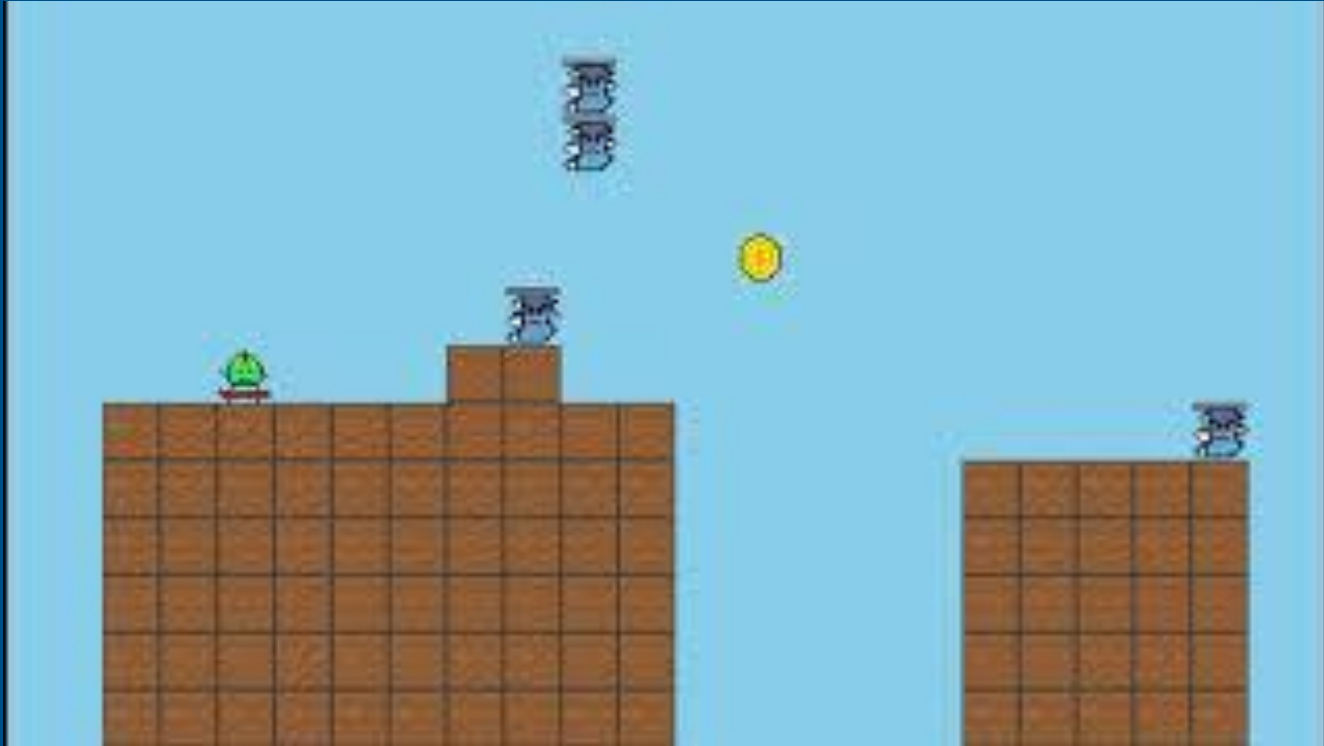
- Poor reward signals
- Environment difficulty
- Delayed rewards



Poor Reward Signalling Example



Impossible Terrain Example



Policy Success Example

